Fault Rotation of Paleomagnetic Directions in Coppermine River Lavas and their Revised Pole

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The paleomagnetic pole of a segment of lavas from the upper part of the Coppermine River Group is displaced relative to that previously obtained for the lower part of the group by a 28-degree, counterclockwise rotation. The inclinations are approximately the same. This segment of lavas is separated from the lower part of the group by a fault zone. Elsewhere where the stratigraphic sequence is continuous, sediments and lavas of the upper part of the group give pole positions, which, although scattered, are in large part close to that of the lower part of the group. Hence, the displaced pole can reasonably be attributed to local rotation along a fault rather than to apparent polar wandering. Recognition of the tectonic rotation of the paleomagnetic pole leads to an interpretation of the faulting in the Coppermine River flows as rift faulting closely following extrusion of the lavas.

New Coppermine River pole positions are calculated combining the data previously obtained from the lower part of the Coppermine River Group with that from lavas (4 sites) and sediments (5 sites) from the upper part of the group. The new Coppermine River pole from the lavas is at 02N, 180E and from lavas and sediments combined is at 03N, 182E (dp = 3, dm = 6). The latter is the pole recommended for use. That this is a Coppermine River pole rather than a Mackenzie pole is emphasized with the thought that the various elements of the Mackenzie magmatic events may ultimately resolve into a succession of discrete episodes.

Le pôle paléomagnétique d'une tranche de laves de la partie supérieure du groupe "Coppermine River" est décalé de 28 degrés dans le sens anti-horaire relativement à celui qui avait été obtenu pour la partie inférieure du groupe. Les inclinaisons sont sensiblement les mêmes. Cette tranche de laves est séparée de la partie inférieure du groupe par une zone faillée. Ailleurs, là où la stratigraphie est continue, les sédiments et les laves de la partie supérieure du groupe donnent des positions polaires rapprochées de celles de la partie inférieure. Par conséquent, le décalage du pôle, peut raisonnablement être imputé à une rotation locale le long d'une faille plutôt qu'à une migration apparente du pôle. Si l'on admet une telle rotation tectonique du pôle paléomagnétique, ceci nous amène à interpréter la faille à l'intérieur des laves du groupe "Coppermine River" comme une faille qui c'est produite immédiatement après l'extrusion de ces laves.

Le calcul des nouvelles positions du pôle du groupe "Coppermine River" est fait en combinant les données déjà obtenues de la partie inférieure du groupe avec celles des laves (4 sites) et des sédiments (5 sites) de la partie supérieure du groupe. Les nouvelles positions du pôle du groupe "Coppermine River" sont 02N, 180E pour les laves, et 03N, 182E (dp = 3, dm = 6) pour les laves et sédiments combinés. Ce dernier est le pôle recommandé pour usage. Ce pôle est celui du groupe "Coppermine River" plutôt que celui du groupe "Mackenzie" parce que les divers éléments de la séquence magmatique du groupe Mackenzie peuvent finalement se ramener à une succession d'épisodes mineurs.

[Traduit par le journal]

Introduction

The original paleomagnetic results from the Coppermine River Group (Robertson 1964, 1969) were derived from the lower part of the sequence only. Subsequently, work by Robertson and Fahrig (1971) summarizing the evidence from other formations demonstrated that the Coppermine River pole was at the threshold of a period of rapid polar movement manifested in the Logan loop. This developed and redefined a pioneer curve outlined by Dubois (1962). Accordingly it seemed desirable to determine if the pole had shifted significantly during the course of deposition of the Coppermine River Group itself. In 1969 a number of samples was collected from red sediments and lavas in the upper part of the group (lettered sites Fig. 5). Many of these results were anom-

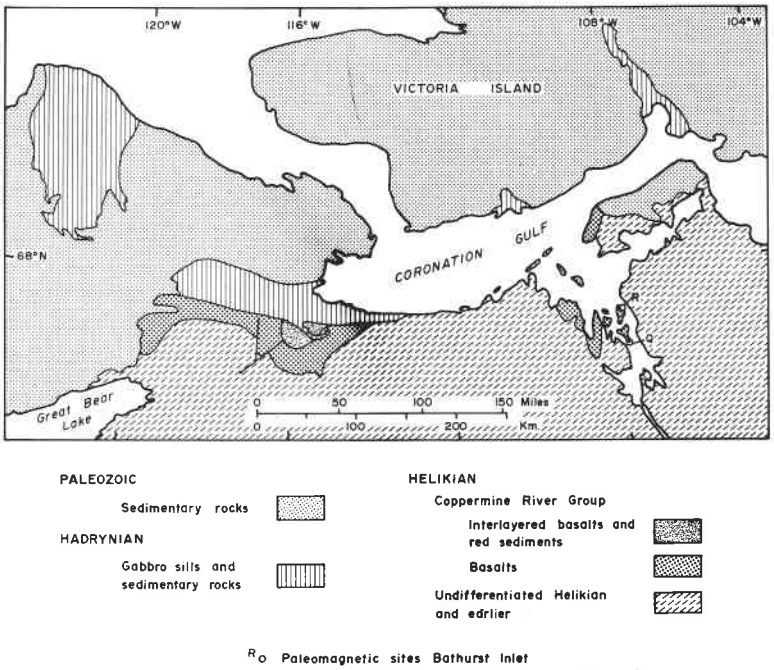


FIG. 1. Index map of Coronation Gulf region showing the distribution of the Coppermine River Group and its geological setting. Q and R are the sites of paleomagnetic samples from Bathurst Inlet.

alous, hence more detailed sampling was done in 1971 in a small area of the upper flows in an attempt to resolve these anomalies (numbered sites Fig. 5). The latest results required a revised structural interpretation of the geology and this forms the core of this paper. It demonstrates the use of paleomagnetism in obtaining solutions to structural problems where the magnitude of relative rotation is required.

Both oriented hand samples, later drilled in the laboratory, and cores oriented in the field were used. Samples were oriented by sun compass and magnetic compass and are considered to be accurate to within 2 degrees. In addition to Coppermine River samples, results from two sites in the Bathurst Inlet area (Fig. 1) are reported for the first time.

General Geology

The distribution and geological setting of the Coppermine River Group is illustrated in Figs. 1 and 2. It forms part of a stratigraphic succession that spans Proterozoic time, as shown in the following table of formations adapted from Baragar and Donaldson (1973).

Several unconformities exist within the succession but, apart from the break that marks the end of the Aphebian, they represent only slight structural discontinuities. All Helikian and Hadrynian strata have dips that are rarely more than 10 degrees. The Dismal Lakes and Coppermine River Groups dip generally northward at mainly 5 to 10 degrees and are overlain by the Rae Group, which also dips northward at 3 to 5 degrees. The unconformity that separates them is conspicuous only because a broad synclinal warp in the Coppermine River Group is sharply truncated by the overlying Rae Group. The synclinal axis protrudes a short distance from beneath the unconformity in the central part of the warp, and several hundred stratigraphic feet of south-dipping basalt flows can be seen to the north of it. The warp and synclinal axis are both critical to an interpretation of our paleomagnetic results and will be discussed in greater detail later.

Two major periods of basaltic magmatism, corresponding to the Mackenzie (Fahrig and Jones 1969) and Franklin (Fahrig *et al.* 1971) magmatic events, are represented in the upper parts of the Proterozoic sequence. Each has

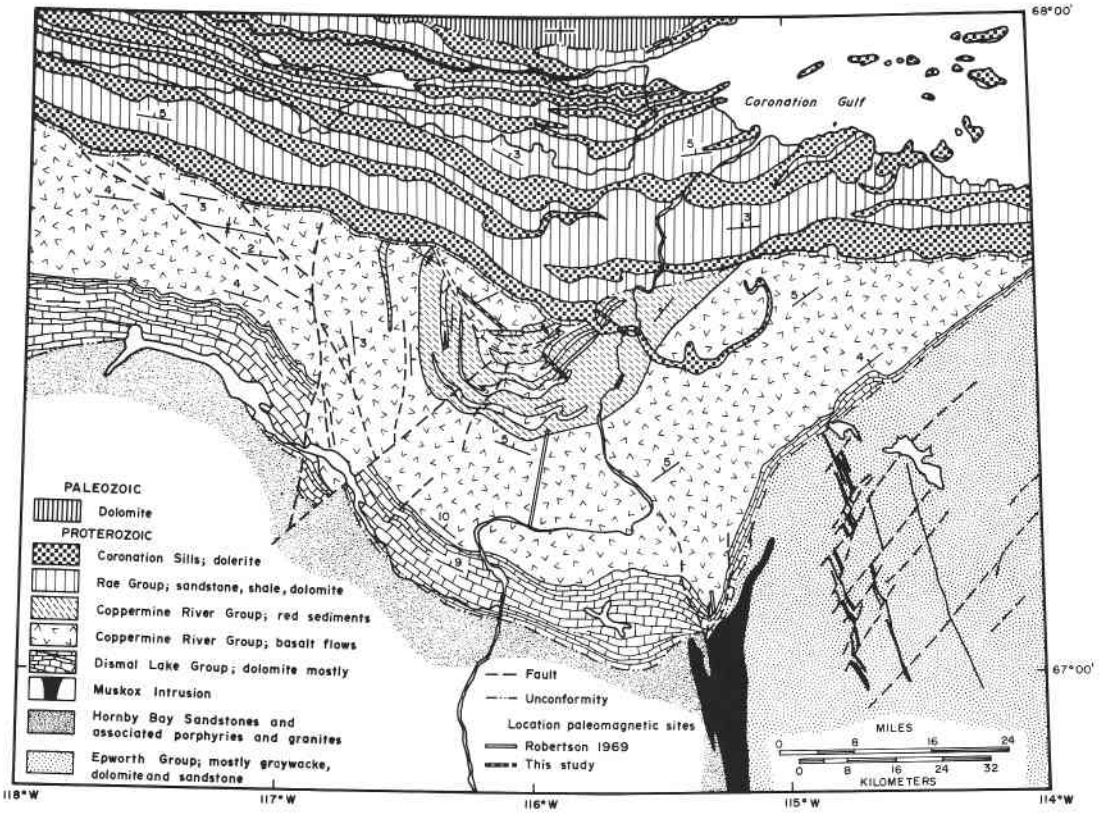


FIG. 2. Geological map of the Coppermine River region showing the location of sampled sections and sites. Adapted from Baragar and Donaldson (1973) with minor modifications.

been characterized by a single paleomagnetic pole, but within this region the Mackenzie events, at least, are complex and a single pole may not be warranted. Elements of the Mackenzie magmatic events are the Mackenzie dikes, the Muskox Complex, and the Coppermine River flows.

The Mackenzie dikes form a diffuse swarm of northwest- to north-northwest-striking diabase dikes that extend from northwestern Ontario to the Coronation Gulf, a distance of about 2400 km (1500 miles). They have been dated by the K-Ar method only and the dates range from 865 to 1360 m.y. (Fahrig and Jones 1969), but their relationship to the Coppermine River flows and the Sudbury dikes (dated at 1205 m.y. (Van Schmus 1964)) led Fahrig and Jones to conclude that they are about 1200 m.y. old. Mackenzie dikes unquestionably feed Coppermine River flows, but dikes with the same trend may be both older and younger than the flows. In Fig. 2, a few of the dikes with a north-northwesterly trend typical of the Mackenzie swarm in this region are shown within the Epworth Group. Some are offset by northeast-trending faults, whereas others are not. One set of thick dikes strikes towards but does not penetrate the Dismal Lakes dolomite. On the other hand dikes with the same trend as Mackenzie dikes are found within the Rae Group, and may be related to the much younger Coronation Sills.

The Muskox Complex was regarded as a fairly direct associate of the Coppermine River flows until recent work by Irvine (1970) demonstrated the existence of a substantial time interval between them. Following intrusion of the Muskox Complex an unconformity developed and was succeeded by nearly 1100 m (about 3500 ft) of dolomites before Coppermine River volcanism began. Nevertheless, the Muskox Complex gives ages (K-Ar 1155± m.y. (Lowdon 1961, p. 22) and Rb-Sr 1189± m.y. (R. K. Wanless, personal communication,

Table of formations

Paleozoic		Dolomite
		unconformity
Proterozoic	Hadrynian	Coronation Sills: gabbro Rae Group: dolomite, quartzite siltstone, shale
		unconformity
	Helikian	Coppermine River Group: Red sandstone and intercalated basalt Basalt Dismal Lakes Group: dolomite, sandstone, mudstone
		unconformity
		Muskox Intrusion: layered ultramafic complex Hornby Bay Group: sandstone,[1] siltstone. and shale
		unconformity
	Aphebian	Granites, quartz feldspar porphyries Epworth Group: graywacke, dolomite, sandstone, siltstone, and shale

[1]A time gap of as much as 400–500 m.y. may separate the deposition of Hornby Bay sediments and emplacement of Muskox Intrusion according to evidence presented by Irving *et al.* (1972) and Park *et al.* (1973). A single paleomagnetic pole from the Hornby Bay Group lies close to the paleomagnetic pole position of the Dubawnt Group, reliably dated at 1725 m.y. Additionally, a minimum age of the Hornby Bay Group is given by a K–Ar age of 1400 m.y. for the Western Channel diabase which cuts it.

1972)) that are broadly similar to the age of Coppermine River lavas, and is probably ancestral to them.

The Coppermine River Group comprises a lower formation of predominantly basalt flows, about 3300 m (10 000 ft) thick, and an upper formation of interlayered basalt flows and red fluviatile sandstones, about 1300 m (4000 ft) thick. Individual flows most commonly range from 8 to 25 m (25 to 75 ft) thick and typically comprise a massive base and an oxidized, amygdaloidal top, generally in the ratio 4:1 to 3:2 respectively. Basalts of the massive parts of flows are generally a little altered; plagioclase is partly replaced by montmorillonite and augite to a minor degree by chlorite. Chlorite pseudomorphs after orthopyroxene(?) and rarely olivine are present in the lowest 650 m (2000 ft) of the lower formation.

The Coppermine River succession records an interval of intense, nearly continuous volcanism followed by a period of declining, intermittent eruptions, during which the accumulation of sediments exceeded that of the intervening flows. Volcanic rocks of the lower formation are related by a common differentiation trend that proceeds upward through the succession. Those of the upper formation do not continue the trend and, therefore, appear to belong to a distinct and later eruptive phase (Baragar 1969).

The indicated age of the Coppermine River flows is about 1200 m.y. (Baragar *in* Wanless and Loveridge 1972). A Rb-Sr isochron of the lower flows gives an age of 1214 ± 45 m.y. ($\lambda = 1.47 \times 10^{-11}\text{yr}^{-1}$) and K-Ar ages from the entire sequence range from 740 to 1200 m.y. There is no systematic change in K-Ar dates with stratigraphic level, and the variation is attributed to differential updating caused by the intrusion of the Coronation Sills.

The Coppermine River Group was warped and faulted before deposition of the Rae Group. The age of the movements is unknown but might reasonably be expected to follow shortly after emplacement. A subsidiary isochron interpreted from three anomalous points on the Rb-Sr plot of Coppermine River lavas gives some support to this view. The samples corresponding to these points are from localities

close to the major north-south fault in Fig. 2, and may have been affected by hydrothermal solutions percolating along the fault zone. If so, the age indicated by the subsidiary isochron (1072 ± 65 m.y.) would represent a minimum age of faulting (Baragar *in* Wanless and Loveridge 1972).

The Rae Group is profusely intruded by the Coronation Sills. These have an average K-Ar age of about 650 m.y. and, as shown by Robertson and Baragar (1972), a paleomagnetic pole position that is consistent with that of the Franklin Magnetic Interval of Fahrig *et al.* (1971). The lowermost sill transects the upper part of the Coppermine River Group (Fig. 2), but may not intrude it. Sediments of the Rae Group are found between the intrusion and the underlying Coppermine River flows and sediments, at a point where the intrusion appears to cross-cut them at right angles (Fig. 5). An alternate explanation is that this part of the intrusion is a remnant of the basal part of a sill that was preserved in an erosional trough on the underlying surface. In either case the sill is close to many of the sites in Group A to P and might well have affected their magnetization directions.

Remanent Magnetization

The magnetization directions from the sediments and lavas at the lettered sites (Fig. 5) are given in Tables 1 and 2. The directions underlined are those that are considered to contain a predominant primary component acquired at the time the lavas cooled and the sediments were laid down. They have been used to compute mean directions, shown in Tables 2 and 4 for comparison with the other results. Directions from the other lettered sites are believed to have been affected by secondary influences, as will be explained in a later section, and were not used. The lavas are represented by only one sample from each site, and are therefore less reliable.

Alternating field demagnetization curves (Fig. 3*a*) for a specimen from each of the sites A to P show a wide spectrum of coercivities, ranging from that at site G, in which two-thirds of the initial intensity remains after cleaning in 1000 oersteds (Oe) to that of site I, in which 90% of the initial intensity is removed in a peak field of 150 Oe. It is noteworthy that site R, from Bathurst Inlet, has characteristics similar to the Coronation Sills (Robertson and Baragar 1972). Sites I and M have large components with low coercivities (Fig. 3*a*), typical of those which have acquired IRM from nearby lightning strikes (Graham 1961). However, sites A, B, and C, which have anomalous magnetization directions, contain notable components in the intermediate coercivity range. Therefore, their anomalous directions may not be accounted for by secondary components of low coercivity.

TABLE 1. Site magnetization directions of Upper Coppermine River lavas, lettered sites

	Directions			
	After 150 A.F. cleaning		After 500 °C thermal cleaning	
Site	*D*	*I*	*D*	*I*
A	284	−41	283	−43
B	268	−61	274	−48
C	266	−60	264	−59
D	258	+12	258	+12
G	261	−2	262	−09
I	161	+71	082	+17
J	266	+09	269	+20
K	241	+11	250	+22
L	260	−44	254	−25
M	341	+22	333	+08

NOTE: Underlined directions are those used for the mean direction given in Table 4.
D = Clockwise declination in degrees from true north.
I = Inclination in degrees; positive downwards.

Thermal demagnetization curves (Fig. 3*b*), obtained from a second specimen of each of the same cores, showed that most sites contain a large proportion of magnetic minerals with blocking temperatures between 500 and 600 °C. Sites L, I, and M, however, contained components with blocking temperatures distributed throughout the 70–600 °C range.

Mean directions for red sediments and lavas of the lettered sites are shown in Tables 2 and 4 respectively, and agree quite well with the mean direction previously determined for the lower flows by Robertson (1969).

Initial magnetization directions of samples from the numbered sites (Fig. 5) were scattered (Table 3). Stepwise demagnetization of pilot

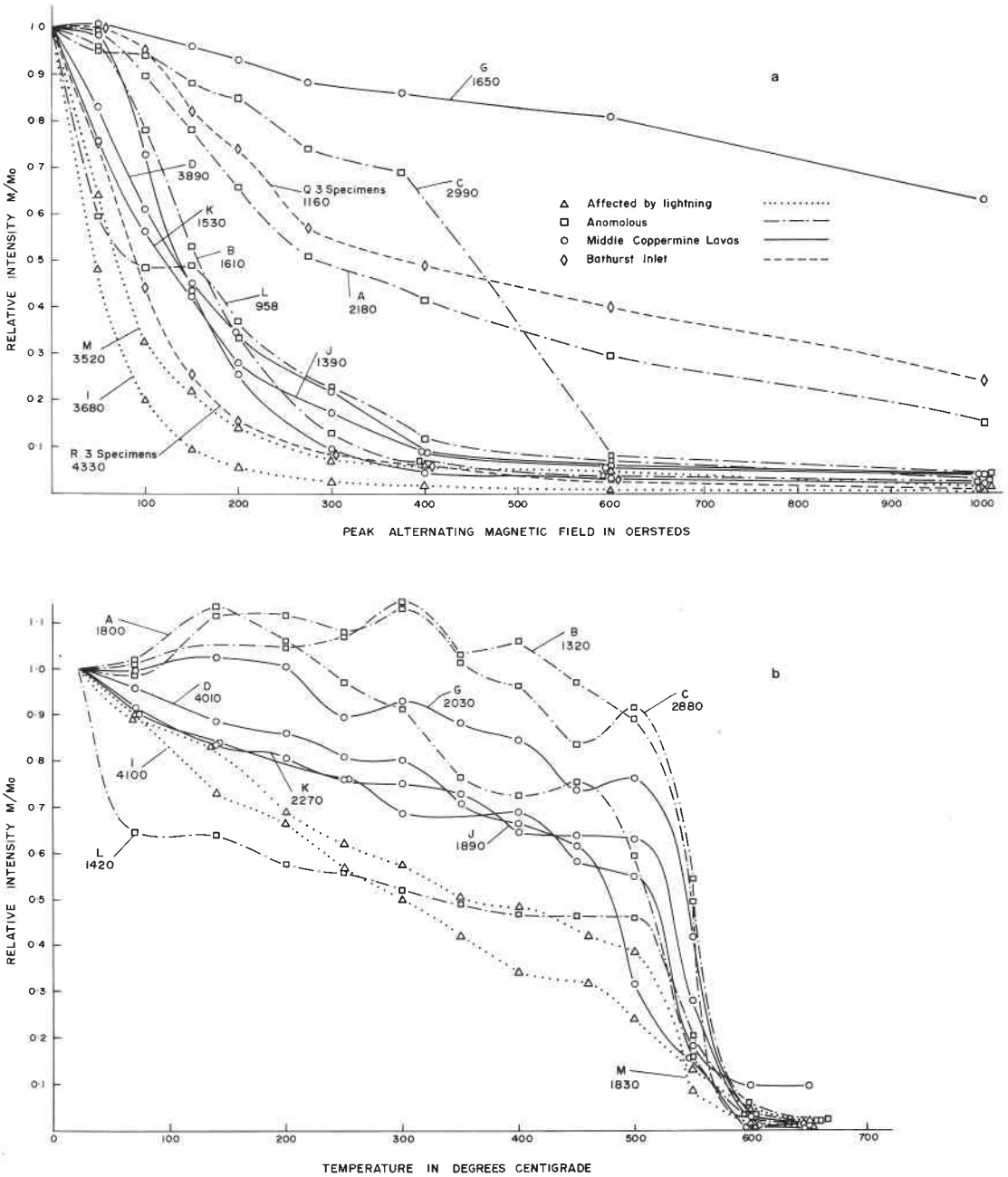


FIG. 3. (*a*) Typical alternating magnetic field demagnetization curves. Middle Coppermine Lavas refer to those samples from the lettered group which are believed to be unaffected by later disturbance and are used in the calculation of the mean pole of the group. Letters refer to sample sites and numbers to initial intensity in e.m.u. $\times 10^{-6}$. (*b*) Thermal demagnetization curves of a second specimen taken from the same sample as those used in 3*a*.

TABLE 2. Coppermine sediments mean site directions and poles after cleaning in 300 Oe peak field

Site	D	I	N	k	α
B	268	−45	4	171	7.1
E	207	+16	2	5.4	—
F	167	+33	2	1.4	—
H	206	+46	3	51	17
N	250	+02	3	14	35
O	250	+34	2	23	55
P	255	+37	3	46	18
Underlined sites	234	+29	5	9	28
Pole position w.r.t. bedding of underlined	Lat. 04	Long. 168W	dp 17	dm 30	

D = Clockwise declination in degrees from true north
I = Inclination in degrees; positive downward
N = The number of samples measured
k = Fisher's (1953) best estimate of precision
α = The half-angle of the cone of confidence at P = 0.95 (Fisher 1953)

specimens from all sites showed that the scatter and inclination decreased in fields up to 150 Oe; in higher fields the scatter increased steadily (although the mean direction did not change appreciably) in fields up to 400 Oe (Fig. 4). Hence, all specimens were cleaned in a peak field of 150 Oe. The site results are shown in Table 3. The within-site dispersion after cleaning is smaller at most sites, but the between-site dispersion is somewhat higher.

The mean direction from the numbered flows (214, +20) is similar in inclination to that of the lower flows (Robertson 1969 and Table 4), but the declination is 28° less. This difference appears from the local geological structures to have been caused by a rotation of the block after it acquired its magnetization.

Mean directions from two sites at Bathurst Inlet are shown in Table 3. That from site R (066, +02) is quite close to the direction for the Coronation Sills (082-04). It seems likely that it is either part of the Franklin Diabases (Fahrig *et al.* 1971), or that it has been baked by them. Site Q is undoubtedly in lava flows, and its direction agrees well with the mean direction of the lower part of the Coppermine River flows. This is good evidence that flows at

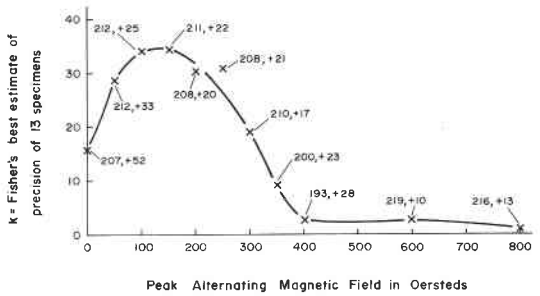


FIG. 4. Precision (grouping) of 13 pilot specimens from sites 1–11 plotted against alternating magnetic field. The numbers give the mean declination and inclination after each cleaning, and do not differ significantly in peak fields ranging from 100 to 350 Oe.

Bathurst Inlet correlate with those of the Coppermine River Group, as has been previously assumed.

Opaque Mineralogy

Polished sections from the numbered sites were examined in conjunction with those of a suite of samples previously taken through the lower formation and volcanic members of the upper formation (Section 1 of Baragar 1969). The object was to determine whether or not the

TABLE 3. Directions and statistics of numbered and Bathurst Inlet sites

	Initial values					Alternating field cleaned values					Pole			
Site	*D*	*I*	*N*	k	α	A.F.	*D*	*I*	k	α	Lat.	Long.	dp	dm
Bathurst Inlet Sites														
Q	252	+27	3	13	36	100	252	+26	19	29	05 N	184 E	21	39
R	050	+09	3	5	60	150	066	+02	128	11	10 S	184 E	5	11
								1971 Sites						
1	244	+63	6	56	9	150	232	+37	322	4	06 N	195 E	3	4
2	204	+39	6	10	22	150	208	+08	35	12	16 S	215 E	6	12
3	232	+64	6	33	12	150	226	+28	24	14	01 S	200 E	8	15
4	223	+57	5	30	14	150	222	+13	5300	1	10 S	201 E	0.5	1
5	208	+31	6	6	31	150	227	−04	11	21	17 S	194 E	10	21
6	222	+77	6	194	5	150	222	+30	242	4	01 S	204 E	3	5
7	216	+53	5	65	10	150	209	+24	939	3	07 S	215 E	1	3
8	214	+52	6	63	9	150	210	+28	503	3	05 S	215 E	2	3
9	199	+27	6	36	11	150	198	+01	201	5	21 S	224 E	2	5
10	200	+27	6	148	6	150	200	+19	371	4	11 S	223 E	2	4
11	215	+56	5	22	17	150	206	+33	210	5	03 S	219 E	3	6
1–11	213	+51	63	12	6	150	214	+20	18	4	08 S	210 E	2	5
1–11	213	+51	11[x]	19	11	150	214	+20	23	10	08 S	210 E	5	10

A.F. peak alternating field
Q = a site of lavas from Kanuyak Island
R = a site of igneous rock from Ekululia Island
11[x] = the number of sites that have been meaned to give the statistics
Other symbols have same significance as in Table 2.

TABLE 4. Mean directions, pole positions, and error parameters

Unit	D	I	k	α	Pole w.r.t. Horizontal Lat.	Pole w.r.t. Horizontal Long.	dp	dm	Bedding	Pole w.r.t. Beds Lat.	Pole w.r.t. Beds Long.
Flows from sites D, G, J, K, (Table 1)											
Cleaned in 150 Oe	256	+08	42	14	02S	168E	7	14			
Cleaned in 500 °C	256	+09	28	18	01S	169E	9	18			
Sediments from sites E, N, H, O, P (Table 2)	234	+29	9	28	02N	193E	17	30			
21 flows lower formation[1]	243	+22	35	5	01N	183E	3	6	270/05N[2]	03N	182E
Muskox 40 sites from central layered series[3]	238	+34	13	6	06N	191E	4	7	270/04N[4]	08N	189E
25 flows[5] and 5 sediments[6]	243	+22	21	6	01N	183E	3	6	270/05N[2]	03N	182E

NOTE: D, I, k, and α have the same meaning as in Table 2. w.r.t. = with respect to.
[1]From Robertson (1969), omitting directions from 3 flows in which sample pairs gave divergent directions. Cleaned in peak field of 75 Oe.
[2]Strike and dip are an average for the entire sequence. Local strike divergences have negligible effect on mean direction of magnetization.
[3]From Robertson (1969, p. 39), cleaned in peak field of 75 Oe.
[4]Computed from 3 bore holes by D.C. Findlay (personal communication 1964).
[5]Results from 21 flows above plus sites D, G, J, K, cleaned in a field of 150 Oe.
[6]Results from 5 underlined sites of Table 2.

sites that give the rotated pole (the numbered sites) are characterized by a distinctive opaque mineralogy. There is very little evidence that they are. In all cases the predominant opaque mineral is magnetite, with very subordinate ilmenite—the ratio is commonly 9:1—and ilmenite is present either as lamellae on 111 planes of magnetite or as marginal rims, globules, or inserts in magnetite. Hematite is present to a varying degree, but in all specimens examined it appears to be secondary; it replaces ilmenite, magnetite, or both. Sulfides were not observed in any of the sections, but very minor native copper is commonly present in the upper part of the lower formation, in flows of the upper formation, and in the numbered sites. The total content of metallics—essentially magnetite, ilmenite, and hematite—increases upward in lavas of the lower formation, from about 2.5% at the base to about 6 or 7% at the top. In flows interlayered with red sediments of the upper formation it is again at a lower level—about 3–4%. Among samples from the numbered sites, however, it ranges from 4 to 7%. This could be interpreted as meaning that the fault block containing the numbered sites is from a level that is equivalent to the upper part of the lower flows, but the evidence is very tenuous.

Interpretation

Magnetization Factors

The magnetization directions from the sediments and lavas at sites A to P (Fig. 5) are quite widely scattered (Tables 1 and 2), in contrast to those of the lower flows, which are quite well grouped ($\alpha_{95} = 6°$ (Robertson 1969)). This scatter may be due to one or more of the following factors. Firstly, the proximity to all of the lettered sites of the Coronation Sill shown in Fig. 2 is an unknown but probably important factor. Because the sill closely parallels the unconformity at the base of the overlying Rae Group, it presumably passed over all the lettered sites at varying but probably no great distances. All sites may have been reheated to a greater or lesser degree, but the underlined directions, being grouped close to the Coppermine River direction, are thought to be the most reliable (least affected). Another close grouping of directions produced by sites A, B, C, and L is puzzling in the context of this explanation, but not impossible. For example, reheating of these four sites to a temperature above the blocking temperature of magnetite, but below that of hematite, could produce consistent intermediate directions, if their mineralogies were similar. These may not necessarily lie along a single great circle joining

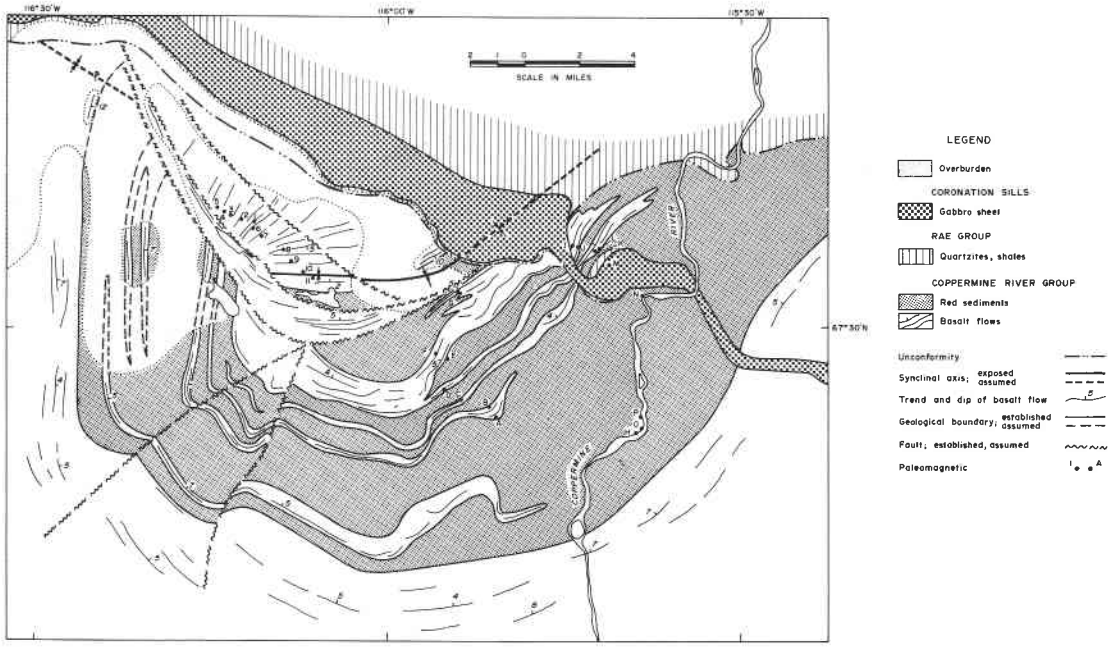


FIG. 5. Detailed geological map of the upper part of the Coppermine River Group in the regions of the synclinal core. Adapted from Baragar and Donaldson (1973) with minor modifications.

the two nearly oppositely directed components (Roy and Robertson 1968). Secondly, some sites (*e.g.* I, M, and R) have acquired a large secondary remanent magnetization, probably from recent viscous build-up or from nearby lightning strikes. Thirdly, sites may have been differentially rotated after they acquired their magnetization. The pole positions of the numbered sites, on the other hand, are relatively well grouped, and accordingly seem to have been little disturbed by local, secondary influences.

Tectonic Factors

The synclinal warp in the upper part of the Coppermine River Group is shown in more detail in Fig. 5. The geology is adapted from Baragar and Donaldson (1973), supplemented by airphoto interpretation and later observations. The numbered sites are all contained within a region bounded on the south by a curving fault zone and on the north by the overlying Rae Group. The synclinal axis that lies in this faulted block swings from an east-west strike in the sampled region near the fault to a northeast strike where it disappears beneath the unconformity. These lavas are assumed to be part of the upper formation, but because of faulting their stratigraphic position is not fully known and they could belong to the upper part of the lower formation, as might be interpreted from their metallic mineral content. Samples from the lettered sites are definitely from the upper part of the Coppermine River Group, and no recognizable faults separate them from the lower part of the group. Their pole positions are scattered, but a significant number of them accord with those previously obtained by Robertson (1964, 1969) from the lower lavas. Therefore, a tectonic rotation of the block containing the numbered sites is indicated. A postulated sequence of events shown in Fig. 6 agrees with the geological observations and explains the 28-degree counterclockwise rotation of the numbered sites. The Coppermine River Group is represented in Fig. 6*A* as a gently folded, gently plunging syncline. In Fig. 6*B* it buckles in response to southward-directed pressure and Fig. 6*C* shows the buckling extended by displacement along a major fault and a series of subsidiary faults that spring from it. One of the subsidiary faults

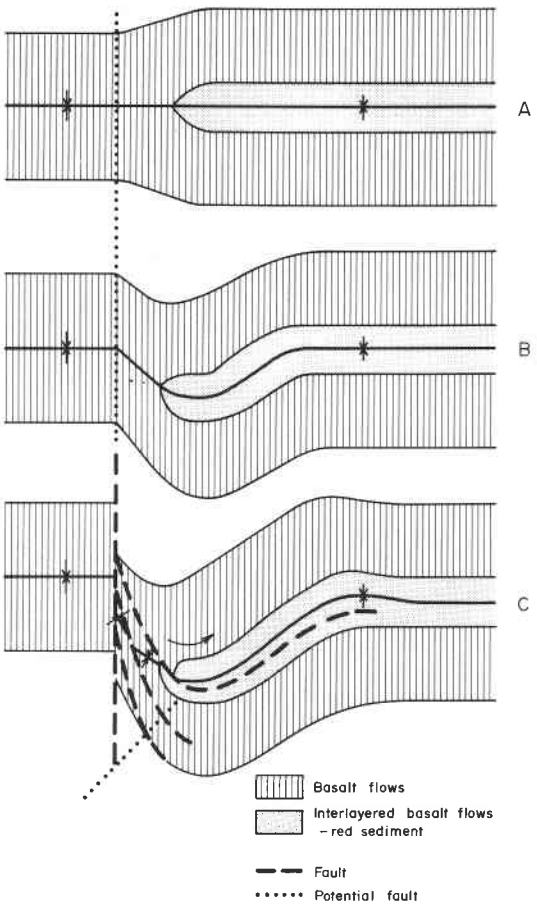


FIG. 6. Schematic diagram showing a succession of events leading to rotational movement of the block containing the numbered sites. (*A*) The Coppermine River Group is gently folded about an east-west axis. (*B*) The group is buckled southward and/or downward. (*C*) Continued movement is taken up on a major north-south fault and several subsidiary faults that spring from it. Movement on the latter is rotational in a counter-clockwise direction.

intersects the synclinal axis, and rotates it and the strata to the north in a counter-clockwise direction.

The force producing the buckle might be gravitational due to withdrawal of magma from below. Although the sketches in Fig. 6 are plan views, they can be regarded as greatly distorted sectional views, as the region has a gentle northward tilt. Even so, some lateral component of force may be necessary, and this can be produced by uneven collapse along the strike of the downwarp. If the collapse is greater to the north the subsidiary faults would have a northerly dip into the deeper part of the collapse. These features are essentially those of a rift zone, which is a structure that would be entirely compatible with the plateau basalts of the Coppermine River Group. The opposing rift fault (Fig. 2) shown as a potential fault in Fig. 6 is a further element of similarity.

Geological observations that are consistent with this interpretation are as follows: (*1*) The Coppermine River Group has a generally east-west strike, as shown by its attitude west of the major north-south fault and by its distribution, as shown in Fig. 1. The southward buckle is an abnormality. (*2*) The Muskox Complex, a concentration of feeder dikes, and a major gravity high all coincide roughly with the axis of the buckle, as might be expected if it represents the source region of the magma. The apex of the gravity high is near Coronation Gulf, north of the surface exposure of the downwarp (Hornal 1967). Irvine and Smith (1967) suggested that this was the source of the magma for the Muskox Complex, and it might reasonably be expected to be the source of the Coppermine River flows as well. If so, it probably marks the site of greatest collapse. (*3*) A major north-south fault and complimentary northeast-southwest fault between them lower a wedge of lavas and sediments in the manner of a rift block (Fig. 2), as assumed in our interpretation. Numerous subsidiary faults of the type depicted in Fig. 6 do seem to originate with the north-south fault, and a few are shown in Fig. 2.

Pole Position

The Coppermine River pole position has been recalculated combining data previously obtained from the lower formation (Robertson 1969) with data from lavas and sediments of the upper part of the group (the lettered sites) obtained in this study see Fig. 7*a*. Only the underlined sites of Tables 1 and 2 were used. The data from all sites have been corrected for a northward dip of 5°, which is a refinement that had not been made in the older data. Two Coppermine River pole positions are presented in Table 4, one containing data from both lavas and sediments (30 sites) and the other, data from the lower lavas alone (21 sites). The positions and associated error ovals

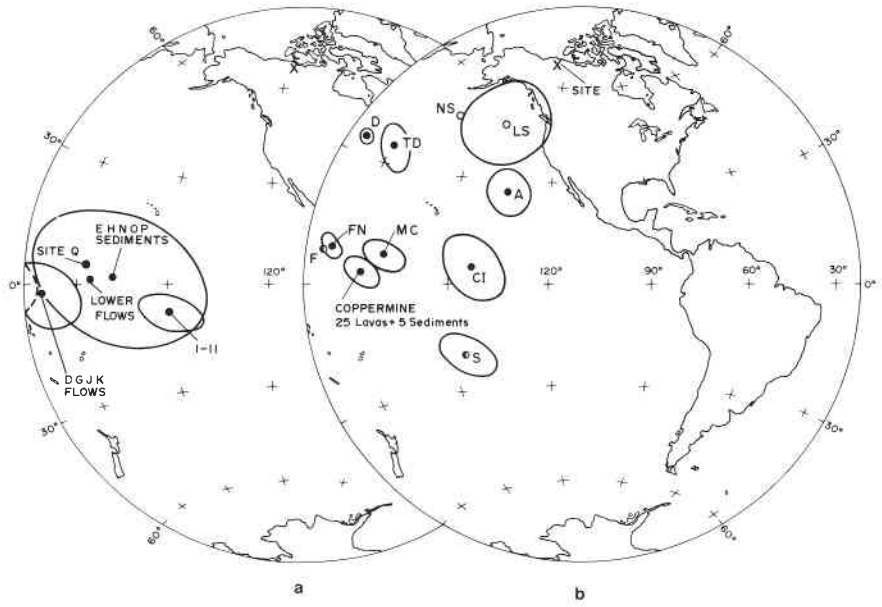


FIG. 7. (*a*) Poles obtained from the new data with their ovals of confidence at P = 0.95, compared with the pole previously obtained from the lower flows (Robertson 1964, 1969). (*b*) The mean Coppermine River paleomagnetic pole with its oval of confidence at P = 0.95 compared with other poles from Proterozoic rocks of the Canadian Shield. In probable order of increasing age the poles are: F = Franklin (Fahrig *et al.* 1971), FN = Freda and Nonesuch shales (DuBois 1962), D = Duluth Gabbro (Beck 1970), TD = Thunder Bay Dikes (Robertson and Fahrig 1971), NS = North Shore Volcanics reversed (Palmer 1970), LS = Logan sills (Robertson and Fahrig 1971), A = Abitibi Dikes (Larochelle 1966), Coppermine River Lavas and Sediments (this report), MC = Muskox central layered series (Robertson 1969), S = Sibley Group (Robertson 1973), CI = Croker Islands (Palmer 1969).

are respectively as follows: 03N, 182E (dp = 3, dm = 6) and 02N, 180E (dp = 3, dm = 6). The pole for combined lavas and sediments is shown in Fig. 7*b*, together with a slightly revised pole for the central layered series of the Muskox Complex (corrected for 4° northward tilt) and other Proterozoic poles of the Canadian Shield for comparison.

Previously (Fahrig and Jones 1969) the data from the Coppermine River flows were combined with data from the Mackenzie and Sudbury dikes and from the Muskox Complex to give a mean pole for the Mackenzie events. The radiogenic evidence that the Coppermine River lavas are the same age as the Muskox Complex and the Mackenzie and Sudbury dikes is not convincing. Dikes with the same trend as the Mackenzie swarm have been shown to be of more than one age and the Muskox Complex is demonstrably older than the Coppermine River flows.

We suggest that greater clarity would result from treating each of these 4 units as separate entities and using the result given here as that for the Coppermine River Group. More detailed sampling may reveal an age difference and, in addition, there is a difference in the reliability of ascertaining the attitude of the rock sampled. Stratified rocks, for example, can be corrected for tectonic tilt, but in most cases dike rocks cannot.

We might consider in this context whether or not the difference in pole positions between the Muskox Complex and the Coppermine River flows can be reasonably explained by their age difference or whether it must be considered an error factor. We have noted that the Muskox Complex is separated from the Coppermine River Group by at least the amount of time necessary to form an unconformity and deposit 1100 m (about 3500 ft) of dolomites—surely not less than 30 to 50 m.y. Using the pole positions for the Coppermine River flows and Muskox Complex shown in Fig. 7*b* this repre-

sents a rate of movement of about 0.3 to 0.5 degrees per m.y. This is similar to the rate of movement of the pole between that of the Logan Sills and the Coppermine River flows (Fig. 7*b*) and to the average rate of polar movement (0.2 to 0.3 degrees per m.y.) for all time determined by Donaldson *et al.* (1973). It seems reasonable, therefore, to separate the Muskox and Coppermine River poles. That is not to say, however, that they can be regarded as separate magmatic events. On the contrary they appear to have close genetic affinities, but the time span represented by their respective emplacements may be significant in terms of polar movement.

Fahrig *et al.* (1971) defined a Mackenzie Magnetic Interval during which a characteristic direction without reversals was imposed upon a large part of the western and central Canadian Shield. The mean age of the interval was said to be about 1200 m.y., but its length was not defined. It spanned a period representing the emplacement of the Coppermine River lavas, the Muskox Intrusion, the Mackenzie dike swarm, and the Sudbury dikes. As we have seen the time spread between at least two of these elements is not negligible. Hence it seems desirable to treat each of these elements separately with a view to resolving their respective poles, if this is possible. In this paper we present separate poles for the Muskox and Coppermine River events (Fig. 7*b*). Ultimately the Mackenzie magmatic events (corresponding to the Mackenzie Magnetic Interval) may be represented by a cluster of poles.

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